Solution Requirements (Functional & Non-functional)

Project Detail	Information
Date	29 October 2026
Team ID / Enrollment No.	EN25CS3010450
Project Name	Global Fuel Prices and Taxation Trends
Maximum Marks	4 Marks
Student	Joyal Sengar (Individual Project)

Functional Requirements:

Following are the functional requirements of the proposed Tableau-based analytical solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
FR-1	Data Access & Integration	Connect Tableau to the global fuel dataset containing date, country, region, petrol price, diesel price, LPG price, Brent crude price, tax percentage, income level and subsidy level.	High
FR-2	Data Processing	Prepare the dataset for analysis by checking data types, missing values, duplicate records, date formatting, category labels and unusual fuel-price values. Keep a record of data-quality decisions.	High
FR-3	Business Analysis	Calculate and compare average petrol, diesel and LPG prices and tax percentages; aggregate results by region, country, income level, subsidy level and date where relevant.	High
FR-4	Business Insights & Analytical Outputs	Provide Tableau worksheets, a dashboard and a story that communicate overall fuel-price measures, country/region comparisons, time-based patterns, tax differences, and income/subsidy comparisons supported by the dataset.	High
FR-5	User Interaction & Analysis	Allow users to read chart titles, labels, units, legends and tooltips and to use any filters or dashboard interactions implemented in the final workbook to explore relevant categories.	Medium
FR-6	Data Update & Reporting	Keep the CSV and packaged Tableau workbook available for review, and export the final dashboard or story to PDF/image format when required for reporting or submission.	Medium

Non-functional Requirements:

Following are the non-functional requirements for usability, data integrity, reliability and maintenance of the project.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Use clear worksheet and dashboard titles, readable labels, consistent number formats and units, and a logical visual layout so users can understand the analysis without extensive explanation.
NFR-2	Data Integrity & Privacy	Use the supplied market-level dataset for analysis and avoid adding unsupported or personal information. Keep the source file and workbook in the intended storage location and share them only as appropriate.
NFR-3	Reliability	Ensure that displayed values correspond to the connected dataset, calculations are checked against source fields, and the workbook opens without missing-data-source errors.
NFR-4	Performance	Keep worksheets and dashboard views reasonably responsive by using appropriate aggregations and avoiding unnecessary visual complexity.
NFR-5	Availability & Backup	Retain the CSV dataset and a packaged Tableau workbook copy so the project can be reopened and reviewed. Online availability depends on where the workbook is published or shared.
NFR-6	Maintainability & Scalability	Use clear field names and documented calculations so the analysis can be updated if new dates, countries or records are added, provided the new data follows the same structure.

Scope note: These requirements are tailored to the supplied fuel dataset and Tableau workbook. Features such as specific filters, online publishing, automatic refresh or sharing permissions should only be marked as implemented if they are actually configured in the final project.